



JOINING THE CHORUS

By howling, wolves advertise their ownership of a hunting territory to any other wolves that may be in the area. Wolves often howl at night, after they have made a successful kill.

different because the signal lingers long after the animal that made it has moved on. Animal scents are specific, allowing animals to lay trails and to advertise their presence to potential mates. Some male insects are able to respond to individual molecules of airborne scent, allowing them to track down females far upwind.

Living in groups

Some animals spend all their lives alone and never encounter another member of their species. But, for many, getting together is an important part of life. Animal groups vary in size as well as in how long they last: mayflies, for example, form mating swarms that last just a few hours, while migrating

GROUP FORMATION

By forming a V (or skein), geese can reduce the amount of energy needed to migrate by flying in the leading birds' slipstream. They take turns to lead.



SAFETY IN NUMBERS

In open habitats, such as grassland, where predators tend to be fast and hiding places scarce, prey mammals typically live in large groups. This group of zebras has further increased its security by grazing near a herd of wildebeest.

birds often assemble for several weeks. Many other animals, including fish and grazing mammals, form groups that are maintained for life. Groups of animals may seem to be easy targets for predators, but the opposite is usually true. Predators find it difficult to single out individuals from a group, so living together gives animals a better chance of survival. Groups are also more difficult to catch by surprise because there is always more than one animal on the alert for signs of danger.

In most animal groups, the members belong to a single species but do not necessarily share the same parents. However, in the most tight-knit

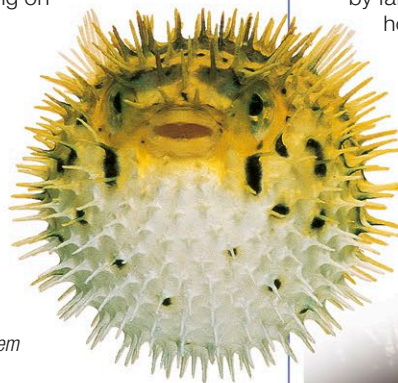
groups, the members are all closely related. Examples of such extended families include wolf packs and kookaburra "clans", where the young remain with their parents instead of setting up independently. This kind of group-living reaches its extreme in social insects, such as termites and ants, which cannot survive alone.

Defence and attack

Both predatory and prey animals use specialized behavior to help them survive. For example, while many prey animals simply try to escape, others keep perfectly still, relying on camouflage to protect them. A wide range of species, from moths to lizards, try to make themselves appear dangerous by

RESPONDING TO THREAT

When threatened, puffer fish enlarge themselves by gulping water. Once distended, they can barely move, but their spines make them practically impossible to attack.



exaggerating their size or by revealing colored spots that look like eyes. Sometimes such threats are real: for example, the brilliant colors of poison-arrow frogs indicate that they contain some of the animal kingdom's most potent poisons.

Predatory animals use one of 2 techniques to catch prey: they either wait for it to come their way or they track it down. "Sit-and-wait" predators are often camouflaged, and some actively entice their victims within range. In anglerfish, for example, the snout has a long, luminous protuberance, called a lure, which the fish dangle appetizingly in front of their mouths: anything swimming close to inspect this lure is snapped up whole. For active hunting, nature puts a premium on speed and keen senses, which is why animals such as cheetahs, peregrine falcons, and blue marlins are among the fastest in the world. Some active predators operate in groups. By working together, gray wolves, African wild dogs, and lions can tackle prey much larger than themselves.

PACK HUNTING

A pack of African wild dogs pulls down a wildebeest that they have run to exhaustion. Once their victim is dead, the dogs will devour it. On returning to their den, they regurgitate some of the meat for any pups that have been left behind.

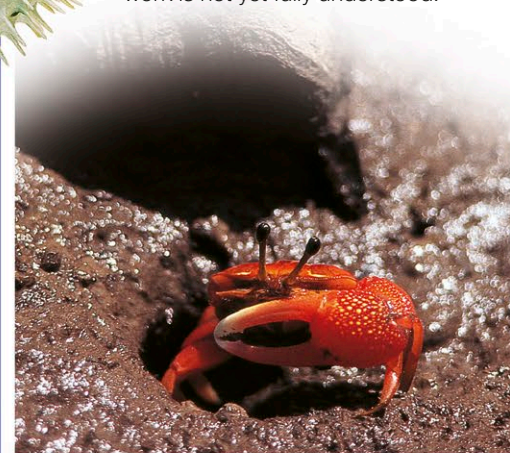


Behavioral cycles

Some kinds of behavior, including self-defense, can be provoked at any time. Others are cyclical, triggered by cues that keep animals in step with changes around them. One of the most important cycles is the alternation between night and day. Others include the rise and fall of the tide and the annual sequence of changing seasons.

Cyclical behaviors are all instinctive. They may be stimulated by external changes, by built-in "biological clocks," or by a combination of the 2. Birds, for example, often gather to roost late in the day, a form of cyclical behavior that is triggered

by falling light levels as the sun nears the horizon. On a much longer time scale, ground squirrels show an annual cycle in body weight, getting heavier before they enter hibernation. However, ground squirrels maintain their cycle even if kept in conditions of constant temperature and day length, which shows that the rhythm is controlled biologically. Biological clocks often involve hormones, but the way they work is not yet fully understood.



FEEDING TIME

Fiddler crabs emerge from their burrows to pick over the nutrient-rich sediment for small particles of food. They cannot feed underwater, so their feeding behavior is governed by the daily rise and fall of the tide.